Nabin Adhikari

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Education

Virginia Tech Blacksburg, VA

M.S. in Computer Engineering, Bradley Dept. of ECE Current GPA: 4.0/4.0 *Jan 2025 – Dec 2026*

- **Graduate Teaching Assistant(GTA) -** Applications of Machine Learning, Introduction to Software Design
- **Relevant Coursework:** Advanced Machine Learning, Foundation Models to AI agents, Introduction to Deep Learning, Trustworthy Machine Learning, Algorithms for Big Data and Nonlinearity & Predictability

Technical Skills

Languages: Python, C, C++, JavaScript, Java, SQL, NoSQL, Solidity, HTML/CSS
Frameworks: PyTorch, TensorFlow, CUDA, Scikit-learn, Hugging Face Transformers, vLLM, llama.cpp, LangChain, LangGraph, AutoGen, LlamaIndex, OpenCV, FastAPI
Tools: Git, Github Actions, Docker, Kubernetes, AWS, MLflow, Airflow, Streamlit, Postman, Jupyter

Experience

AI for Technology Licensing & Discovery Intern Mar 2025 – Aug 2026

Virginia Tech – Office of Research & Innovation *Blacksburg, VA*

- Created 270+ AI-assisted patent marketing briefs supporting patent filing, licensing, and commercialization.
- Improved Tech Publisher platform using AI based models and patent intelligence systems to streamline workflows.

Software Engineering Intern May 2025 – Aug 2025

Virginia Tech – Vet Med Research & Graduate Studies *Blacksburg, VA*

- Enhanced departmental web pages and implemented automated PDF formatting tools.
- Managed CMS validation, content updates, and internal web development tasks.

Co-founder & CEO Sep 2022 – Jan 2025

Laxmi Soft Pvt. Ltd. *Kathmandu, Nepal*

- Launched two enterprise systems: **Laxminn**(50+ hotels) and **SchoolsIE**, a school management platform adopted by a major school in Nepal.
- Led AI-based enterprise solution design, team building, and strategic growth initiatives.

Founding Software Engineer Aug 2022 – May 2023

BuildSupply (MIT Funded) *Boston, MA*

- Designed and built the core AI-powered e-commerce platform end-to-end, developing scalable backend services, databases, and APIs to support SMB-focused building supply procurement.
- Developed and integrated AI/ML components to optimize supplier discovery, pricing efficiency, and user experience.

Projects

Temporal Tree Canopy Correction with Geospatial Deep Learning Github

Python, PyTorch, U-Net, Prithvi ViT, LiDAR, Raster Processing

- Developed a spatial deep-learning pipeline to correct 30 m tree-canopy estimates using environmental and LiDAR data, achieving 25.76 RMSE and 0.686 R^2 while outperforming Random Forest and the NLCD baseline.

Autonomous Hardening of LLM Defenses via Adversarial Feedback Github

Python, PyTorch, Hugging Face Transformers, PEFT (LoRA), Llama 3.1 (8B/70B), DeBERTa, WandB, NVIDIA H200

- Built a closed-loop adversarial system to detect and patch LLM vulnerabilities in real time. Reduced exposure to unseen prompt attacks by approximately 22%.

Time-Aware U-Net Image Generator Github

Python, TensorFlow, Keras, Oxford-IIIT Pet dataset

- Designed a custom time-aware U-Net with ResNet blocks and self-attention for image generation on Oxford-IIIT Pet dataset.

Multimodal Emotion Classification Github

Python, PyTorch, BERT, ResNet

- Developed a multimodal emotion recognition system combining video, audio, and text using pre-trained BERT and ResNet models.

Involvements

- Review Committee for LAUNCH Proof of Concept Grant ($50k funding to support translational research)
- Completed 10+ certifications in AI, ML, Deep Learning, and Applied Mathematics (MIT, Harvard, Coursera).
- Publications: Over Speed Surveillance System Using Deep Learning and Distributed System (The Journal of Engineering, under review) and Towards Energy Equity: Technological Interventions for Reliable Electricity in the Himalayas (Next Energy, under review).
- Course Instructor for *"Data Science and Machine Learning in Python"* at Deerwalk Institute of Technology.
- President, CS Club (1 year) — led technical events, workshops, and student initiatives.
- Speaker on *"AI and Blockchain"* at Esatya.
- Winner, DeerHack Hackathon (National-level Hackathon).